

Enterprise Production Network Infrastructure

Technical Deployment & Architecture Documentation

1. PROJECT OVERVIEW

This document details the high-availability, multi-site enterprise network infrastructure designed and simulated via GNS3. The architecture connects two primary corporate sites (**Cairo Head Office** and **Alexandria Branch**) via a secure, redundant multi-provider ISP Core backbone, ensuring seamless internet availability, secure inter-branch data transport, and centralized resource management.

2. CORE TECHNICAL ARCHITECTURE & SECURITY FEATURES

Routing & WAN Optimization

Basic BGP

OSPF v2

DMVPN (Dynamic Multipoint VPN)

Inter-VLAN Routing

NAT (Network Address Translation)

High Availability & Redundancy (Layer 2/3)

HSRP / VRRP (Virtual IP: 192.168.x.100)

Layer 3 EtherChannel (Bundled Link Aggregation)

Network Hardening & Infrastructure Protection

Port Security (MAC-based Control)

STP Root Guard

STP BPDU Guard

Secure Management (SSH & ACL-bound Telnet)

Centralized Identity & Domain Services

Windows Server AD Domain Controller

Centralized DHCP Server Scopes

3. IP ADDRESSING & SUBNET ARCHITECTURE

Cairo Site (HQ)

VLAN	SUBNET	PURPOSE
VLAN 10	192.168.10.0/24	Windows Server & Core Users
VLAN 20	192.168.20.0/24	Department B Users
VLAN 30	192.168.30.0/24	Department C Users

Alexandria Site (Branch)

VLAN	SUBNET	PURPOSE
VLAN 50	192.168.55.0/24	Branch Core Users
VLAN 60	192.168.60.0/24	Branch Dept 2
VLAN 70	192.168.70.0/24	Branch Dept 3
VLAN 80	192.168.80.0/24	Branch Dept 4

VLAN	SUBNET	PURPOSE
VLAN 40	192.168.40.0/24	Department D Users

WAN Backbone Point-to-Point Interconnections

LINK SEGMENT	SUBNET RANGE	ROUTING PROTOCOLS / PEERING DETAILS
Cairo Edge (R1) to ISP1	10.0.0.0/24	Static/BGP Edge Hand-off Interconnection
Alexandria Edge (R2) to ISP4	20.0.0.0/24	Static/BGP Edge Hand-off Interconnection
ISP Core Internal Mesh Links	30.0.0.0/24 • 40.0.0.0/24 • 50.0.0.0/24 • 60.0.0.0/24	iBGP Peering Mesh overlaid with EIGRP Core Routing

4. TRAFFIC FLOW OPERATIONAL PROCESS

- 1. Endpoint Boot & Addressing:** Host units initiate DHCP discover broadcasts. Centralized **Windows Server DHCP Relay** dynamic scopes issue configuration bindings securely based on incoming VLAN IDs.
- 2. Layer 2 Access Protection: Port Security** instantly monitors incoming frames. Unregistered MAC addresses breach the profile causing immediate automated port mitigation. Simultaneously, **STP BPDU/Root Guards** isolate rogue switch inserts to prevent tree convergence manipulation.
- 3. Resilient Default Gateways:** Inter-VLAN traffic passes up to redundant multi-chassis architectures. **HSRP/VRRP Virtual VIP (192.168.x.100)** presents an active path, guaranteeing instantaneous sub-second failover if a primary gateway uplinks fail. High-speed switch cross-connect aggregates are established via **Layer 3 EtherChannels** to prevent processing bottlenecks.
- 4. WAN Tunneling & Cloud Access:** Inter-site corporate traffic dynamically triggers mGRE encapsulation via secure **DMVPN with OSPF** routing directly across the WAN cloud. General Internet traffic triggers dynamic address translation via **NAT** at edge boundaries, routed across public segments using optimized path vectors via **BGP peering**.

Design Verdict: Fully redundant, resilient, enterprise-compliant architecture blueprint successfully eliminating all single-points-of-failure (SPOF) while maximizing access security.